

GenoSuite: a self-contained desktop application for numerical-genomics analytics

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

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Editor: 

Submitted: 25 June 2026

Published: unpublished

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Summary

GenoSuite is a free and open-source desktop application for the multivariate and population-genetic analysis of genetic marker data. Through a single point-and-click interface it computes genetic distance and similarity matrices, hierarchical clustering and neighbour-joining trees, ordination by principal component analysis (PCA) and principal coordinates analysis (PCoA), the Mantel matrix-correlation test, diversity and population-differentiation statistics (minor-allele frequency, heterozygosity, polymorphism information content, and Nei's F_{ST}), the genomic relationship matrix, linkage disequilibrium, genome-wide association scans, and cross-validated genomic prediction. GenoSuite is implemented in R (R Core Team, 2024) with the Shiny framework (Chang et al., 2024) and is distributed as a self-contained Windows installer that bundles R and all dependencies, so end users run it without installing R or writing code. It is aimed at students, breeders, and biologists who need reproducible analyses of single-nucleotide polymorphism (SNP) and other marker data.

Statement of need

Analysis of marker data typically proceeds through a standard sequence of multivariate and population-genetic steps. The classical workflow—distance matrices, clustering, ordination, and matrix comparison—has long been served by graphical tools, while contemporary quantitative-genomics methods such as the genomic relationship matrix, genome-wide association studies (GWAS), and genomic prediction are available chiefly as R or command-line packages that require programming. Many experimentalists and breeders therefore cannot easily apply modern methods, or must combine several disparate tools.

GenoSuite brings both the classical and the contemporary methods into one graphical workflow that requires no programming or proprietary software:

- **Distance and clustering.** Euclidean, Manhattan, correlation-based, Jaccard, and Gower distances; hierarchical clustering by UPGMA, Ward, complete, and single linkage; and neighbour-joining trees with export to Newick format (Paradis & Schliep, 2019).
- **Ordination and matrix comparison.** Principal component analysis on the markers and principal coordinates analysis on the distance matrix, each with scree plots, and the Mantel test between two distance matrices (Oksanen & others, 2022).
- **Diversity and population structure.** Per-marker minor-allele frequency, observed and expected heterozygosity, polymorphism information content, and population differentiation by Nei's F_{ST} (Nei, 1973).
- **Kinship and linkage.** The genomic relationship matrix following VanRaden (2008), and pairwise linkage disequilibrium (r^2) with a decay plot.
- **Association and prediction.** Single-marker GWAS with principal-component correction for population structure, and cross-validated genomic prediction—genomic best linear unbiased prediction (Endelman, 2011; Meuwissen et al., 2001) together with penalised

regression, random forests, gradient boosting, and a stacked ensemble—reporting predictive ability and genomic estimated breeding values.

Analyses accept comma-separated or Excel marker tables and produce interactive figures and exportable results (distance matrices, trees, statistics, and breeding values).

State of the field

Graphical, install-and-run tools for marker-data analysis are dominated by the proprietary NTSYS-pc (Rohlf, 2000), which remains widely used for distance-based clustering and ordination but is closed-source, no longer actively developed, and predates the genomic era—it offers no genomic relationship matrix, GWAS, or genomic prediction. The spreadsheet add-in GenAlEx (Peakall & Smouse, 2012) provides population-genetic summaries within Microsoft Excel, and the free Java application TASSEL (Bradbury et al., 2007) performs association and diversity analyses but focuses on association mapping and does not cover numerical-taxonomy workflows or genomic prediction. Contemporary prediction and mixed-model methods are otherwise available mainly as R or command-line packages aimed at programmers.

GenoSuite complements these tools by consolidating classical numerical-taxonomy workflows and modern quantitative-genomics methods in a single free graphical application, and by shipping as a self-contained installer that bundles its own R runtime so it can be installed and run without any programming environment.

Software design

GenoSuite is organised as a set of independent analysis modules that share a common data model—a table of individuals (rows) by markers (columns), with optional grouping and phenotype columns. Each module reads this shared input and produces interactive figures and exportable results. Numerical routines build on base R, ape (Paradis & Schliep, 2019), vegan (Oksanen & others, 2022), and the companion package GSBench for genomic prediction, while the graphical interface is built with Shiny (Chang et al., 2024) and bslib. For distribution, the application is assembled into a relocatable bundle—a private copy of R (R Core Team, 2024), all required packages, the application, and a launcher—and compiled into a single Windows installer, so it runs as a standalone desktop application without a separate R installation. The codebase includes an automated test suite covering the analysis and prediction functions.

Research impact statement

By lowering the technical barrier to standard genomic analyses, GenoSuite is intended to support teaching and applied plant- and animal-breeding and conservation programmes, where reproducible analyses are needed but programming expertise or licences for proprietary software may be unavailable. Bringing distance and diversity analysis together with GWAS and genomic prediction in one reproducible interface allows users to move from raw marker tables to selection decisions without switching tools.

AI usage disclosure

Anthropic's Claude (Claude Opus 4.x, via the Claude Code assistant) was used to assist with the following: implementing the application and packaging code, drafting the documentation and user manual, and drafting an initial version of this paper. I defined the scope and selected the statistical methods, directed the development, and reviewed, edited, tested, and validated all AI-assisted output—for example, checking the GBLUP solver against the rrBLUP package.

84 All implemented methods are standard, published techniques in quantitative genetics. I made
85 all final design decisions and accept full responsibility for the work.

86 Acknowledgements

87 I acknowledge the R community and the maintainers of the open-source packages on which
88 GenoSuite depends.

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